

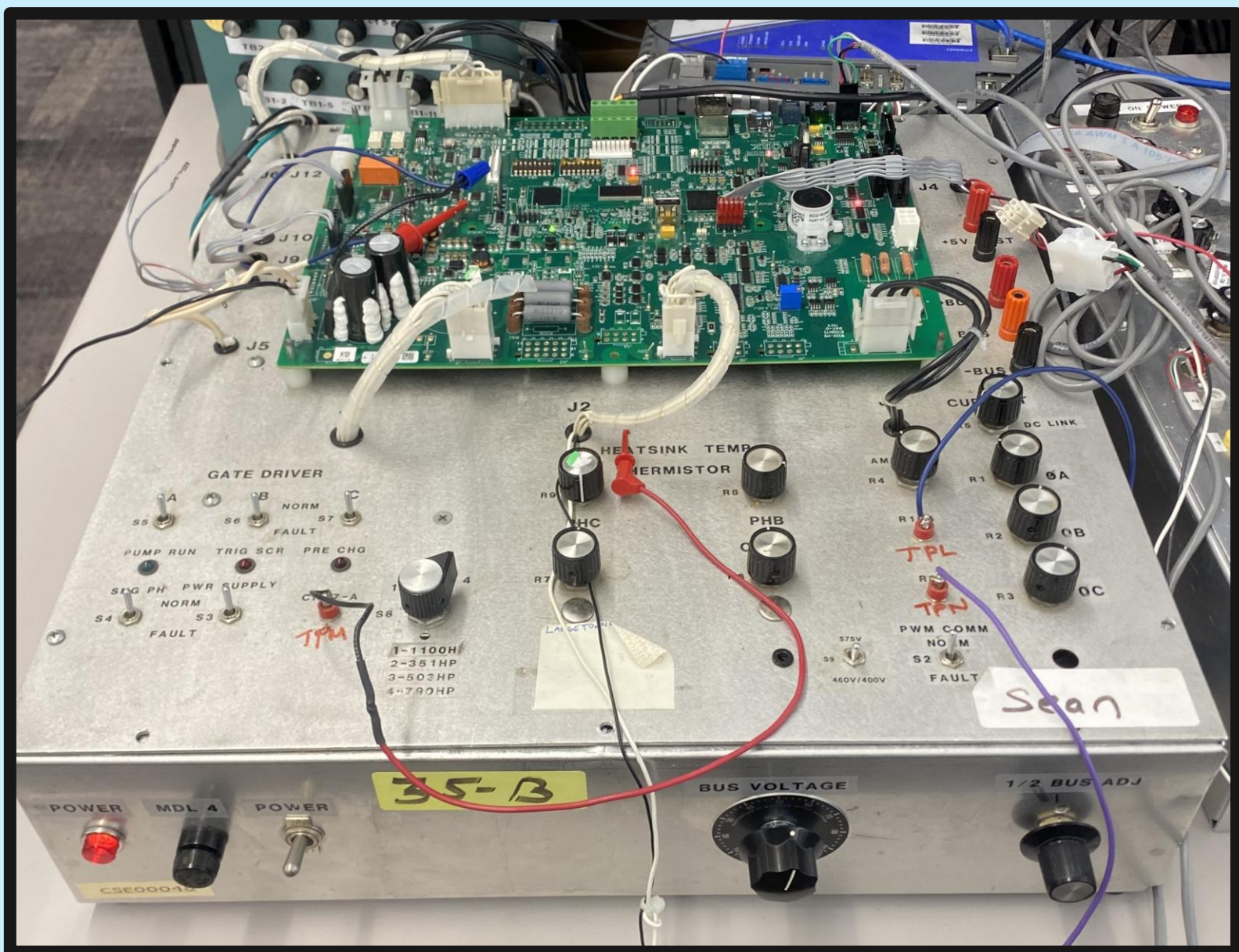
Variable Speed Drive (VSD) Software Test Automation – Brandon Woodward

Spring 2025 | CS490 | Supervisor: Dr. Joe Villani | Faculty Advisor: Dr. David Babcock

I worked for Johnson Controls Inc over the Spring 2025 semester. Supervised by Dr. Villani, I was tasked with testing the VSD 4506 that is embedded within large chiller systems. First, I investigated the test case repository; I found 26 test cases that were applicable. I was able to correctly test 15 of them initially. Five of them were for a specific version of the board I did not have. Following the test case instruction, five of them I was unable to complete. Another four were unclear. After presenting my findings, I worked with the writers of the test cases to update documentation and increase clarity. Then began my second task, automation of these test cases. Using LabView and a special test box, I was able to automate the test cases by communicating with Data Acquisition Devices, Modbus, and Recommended Standard 232. I created the foundations of a software test suite that will be adopted and expanded upon by a future employee, who will refer to the requirements document created by me and my mentors. Successfully implementing 14 test cases and creating core functionality. With the end goal being to test the VSD software system with one simple button click, “Test All”.

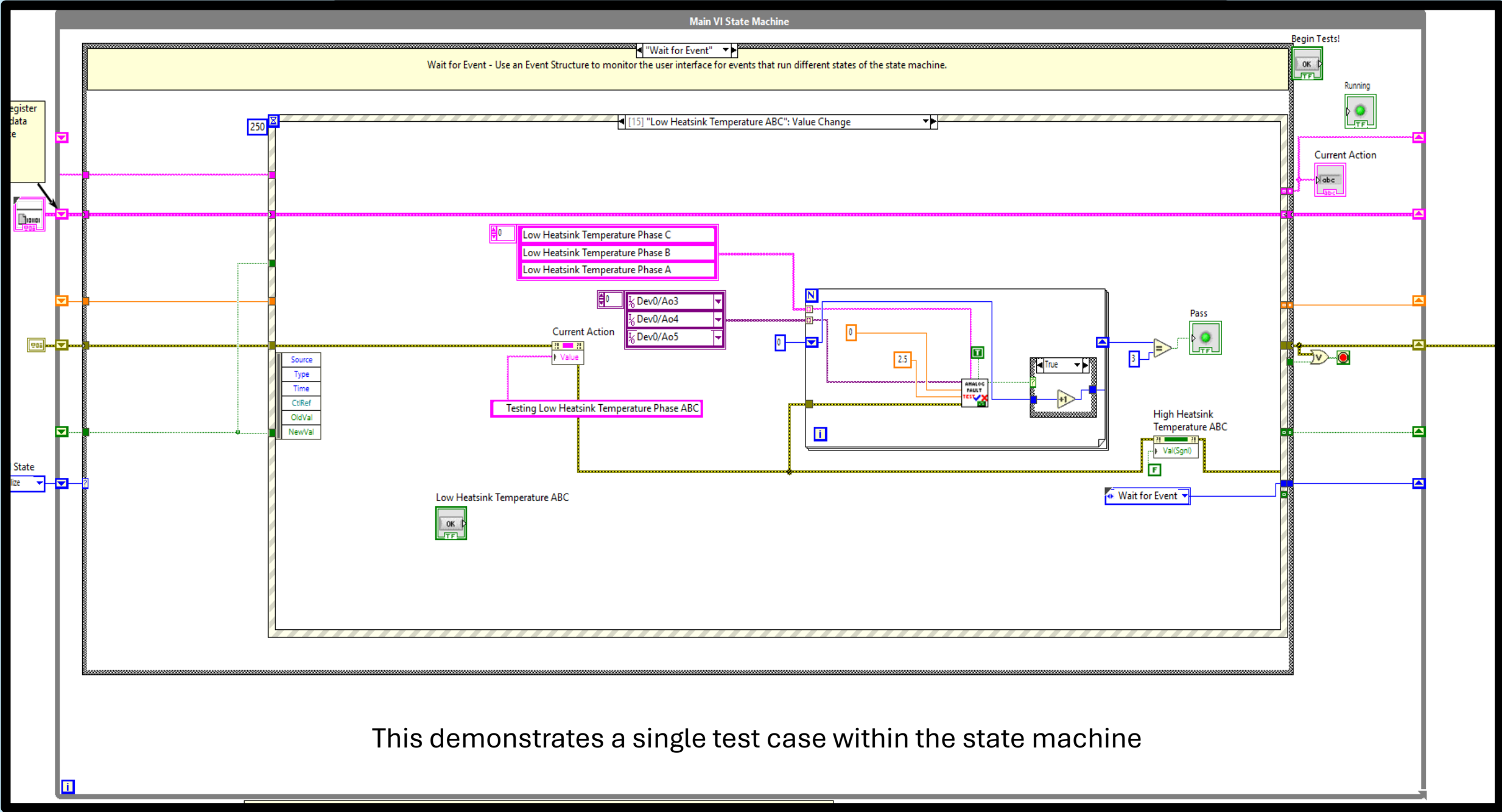


Old Test Box (below): Where I began my testing investigation



Panel installed on all chillers distributed by Johnson Controls (above).

- LabView creates a frontend and backend simultaneously, allowing for rapid development of functionality and user interface.
- Used LabView to emulate the Panel communications via Modbus and Data Acquisition devices.
- Created a state machine to iterate through test cases.



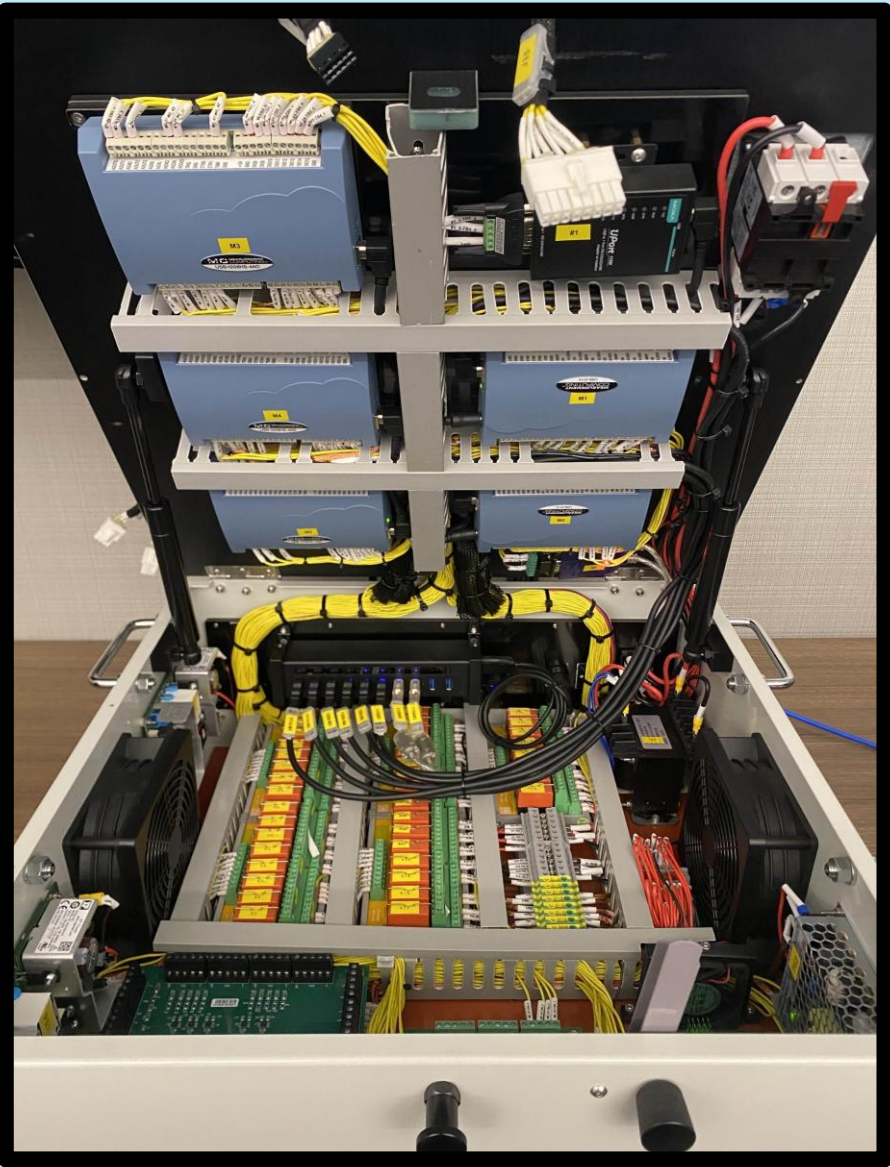
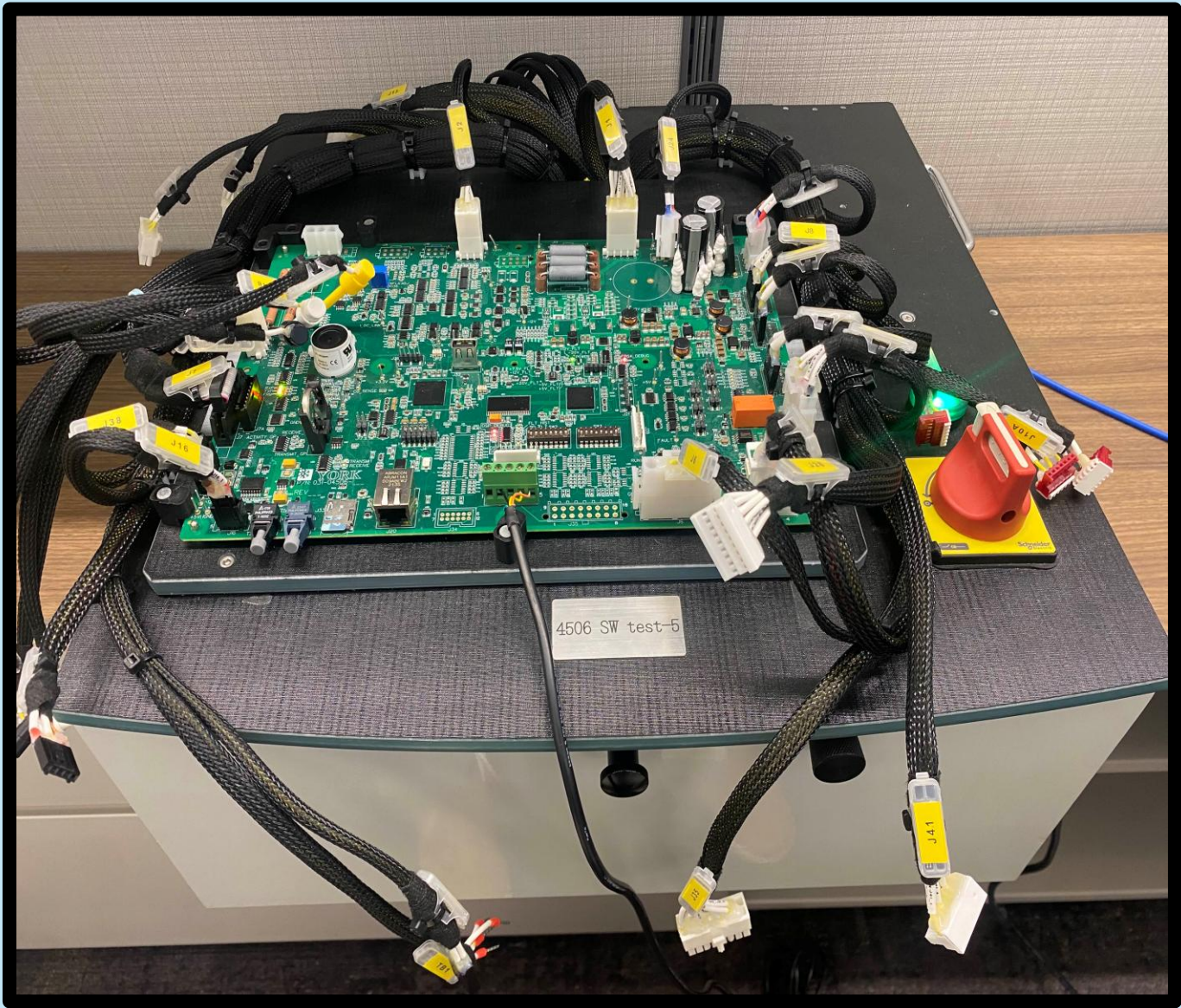
This demonstrates a single test case within the state machine

Relevance:

- Reduces Labor Costs: Eliminates the need for manual test execution, saving time and personnel resources.
- Improves Accuracy: Removes human error, ensuring consistent and reliable test results.
- Ensures Precision: Delivers exact timing and execution sequences critical for embedded systems validation.
- Accelerates Development: Enables faster iteration and integration of VSD software updates through rapid, repeatable testing.

What I learned:

- The power of testing a system before beginning to work with it. One can learn much about it and it helps greatly with the future work.
- Thoroughly reading documentation provides an incredible amount of insight into a system and ultimately allows one to avoid a multitude of questions.
- Full test coverage is a vital component of any large project. Creating and automating these tests is an extremely valuable skill that can set an individual apart.
- To take full advantage of the people around you. They can only accelerate your growth and learning rate. If you have the right questions and are prepared to learn, people are excited to provide you with guidance. Especially at Johnson Controls.



This box has the VSD board mounted on top of it. The blue boxes inside of the box allow us to send and read data directly from the jumpers we have connected to the board. Along with the Modbus, this is how we communicate with the board.



This is the automated testing suite I created for the 4506. It can test all tests with one click, or you can manually choose tests.